

Scalar and Vector Field:

A scalar quantity given as a function of space and time is called a **Scalar field**. Pressure, Temperature, and Density are scalar quantities.

$$p = p(x, y, z, t) = p(r, \theta, z, t) = p(r, \theta, \Phi, t)$$

$$T = T(x, y, z, t) = T(r, \theta, z, t) = T(r, \theta, \Phi, t)$$

$$\rho = \rho(x, y, z, t) = \rho(r, \theta, z, t) = \rho(r, \theta, \Phi, t)$$

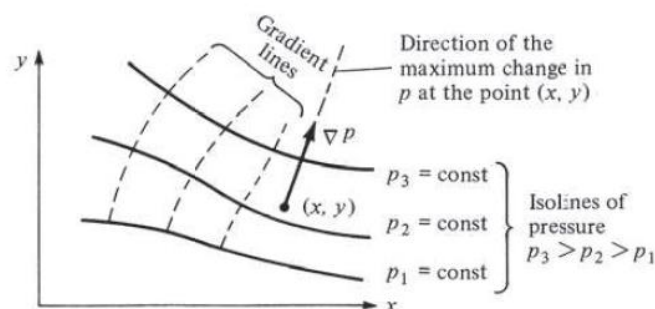
A vector quantity given as a function of space and time is called a **Vector field**. Pressure, Temperature, and Density are scalar quantities. Velocity is a vector quantity.

$$\vec{V} = \vec{V}(x, y, z, t)$$

Gradient of a scalar field:

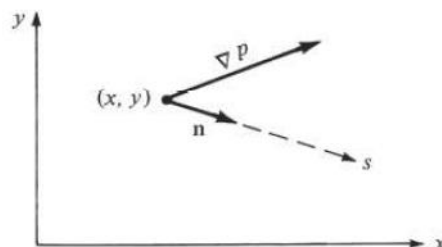
The gradient of any scalar quantity (for example **pressure, p**) at a given point in space is defined as a vector such that:

1. **Gradient** magnitude is the maximum rate of change of p per unit length of the coordinate space at the given point.
2. **Gradient** direction is that of the maximum rate of change of p at the given point.



Isoline – the line along which the pressure remains constant. Moving from one isoline to another, pressure changes.

Gradient line – line along which the gradient of a scalar is tangent at every point.



Choose some arbitrary direction **s** away from the point (x, y) :

Let **n** be a unit vector along the direction **s**. The rate of change of **p** per unit length in the **s** direction is

$$\frac{dp}{ds} = \nabla p \bullet \vec{n}$$

The rate of change of p in any arbitrary direction is simply the component of ∇p in that direction.